

Hybrid Information Retrieval for Tip-of-the-Tongue Dataset

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Abstract—In this research, the impact of retrieval techniques such as lexical, dense, and hybrid models is investigated on the Tip-of-the-Tongue (ToT) dataset. The BM25 model is used for lexical retrieval. SBERT and BGE-M3 are used as dense retrieval models. In addition, hybrid approaches combining lexical and dense models are evaluated. To analyze model performance across different query groups. All models are evaluated using standard information retrieval metrics, including Recall@K, Reciprocal Rank (RR), and nDCG@K.

Keywords—BM25, SBERT, BGE-M3, MonoT5, Hybrid Retrieval, Recall@K, RR, nDCG@K

I. INTRODUCTION

The Information Retrieval (IR) systems strive to extract relevant information from vast document repositories in response to user queries. Nevertheless, user queries may be ambiguous and vague. One instance of query ambiguity is the Tip-of-the-Tongue (ToT) phenomenon when users recall the target information partly but fail to specify their queries accurately. As a result, vocabulary mismatches become common and hamper the effectiveness of lexical retrieval models, such as BM25.

A series of recent works [1] shows that dense embedding models, like SBERT, can handle semantically ambiguous queries by leveraging the contextual semantic similarity of embeddings. Furthermore, recent research [2] has shown that powerful neural reranking models, such as MonoT5, can enhance retrieval performance in hybrid retrieval systems. Moreover, advanced dense retrieval models, including BGE-M3 [3], combine lexical and semantic retrieval signals within a unified retrieval framework.

In our experiments, we assess the performance of lexical, dense, hybrid, and neural reranking retrieval models on the TREC 2024 Tip-of-the-Tongue retrieval task. We consider the following retrieval systems: BM25, SBERT, BGE-M3, MonoT5, and BGE-Reranker. The retrieval systems will be evaluated by Recall@k, RR, and nDCG@k measures.

II. LITERATURE REVIEW

Conventional information retrieval systems largely depend on lexical retrieval approaches like BM25. Study [4] notes that BM25 evaluates documents' relevance using frequency statistics of the terms contained in them. While BM25 is good at retrieving lexically similar documents, it usually lacks the

ability to retrieve semantically similar documents when users query for them. This is highly significant in the case of the Tip-of-the-Tongue (ToT) retrieval problem since user queries are usually incomplete and semantically ambiguous.

As a solution to this problem, there have been some dense retrieval models developed using neural embeddings. For instance, SBERT [1] enhances semantic retrieval using contextualized sentence embeddings, whereas recent neural dense models, such as BGE-M3 [3], combine both semantic and lexical retrieval cues into one framework. SBERT and BGE-M3 are chosen in this research due to their excellent abilities in semantic retrieval.

Apart from dense retrieval, current research findings also show that hybrid retrieval systems utilizing both lexical and semantic signals in retrieving documents achieve significant enhancements in retrieval performance [2]. Accordingly, hybrid retrieval pipelines incorporating BM25 and dense retrieval scores through weighted interpolation will be examined in this research. Moreover, the use of MonoT5 [2] as a neural reranking approach will be considered to determine whether interactive reranking would add more value to the hybrid retrieval process.

Regarding the retrieval mechanism, BM25 retrieval requires minimal preprocessing steps such as lowercase conversion, punctuation elimination, and stopwords removal. Nevertheless, for dense retrieval methods, textual input is not preprocessed as it heavily depends on the semantic nature of texts.

III. RESEARCH QUESTIONS AND METHODOLOGY

A. Research Questions

RQ1 (Lexical-Semantic Gap): To what extent do zero-shot dense retrieval models, SBERT and BGE-M3, improve retrieval effectiveness over the BM25 lexical baseline for Tip-of-the-Tongue queries?

RQ2 (Hybrid Synergy): Does weighted interpolation of BM25 and dense retrieval scores produce statistically significant gains over standalone lexical and dense retrieval approaches?

RQ3 (Reranking vs. Hybrid Retrieval): Under the same BM25 candidate pool, how do linear hybrid retrieval methods compare with neural reranking models such as MonoT5 and BGE-Reranker in terms of retrieval effectiveness and com-

putational efficiency measured by nDCG@10, RR@10 and Recall@10?

B. Proposed Architecture

The proposed retrieval framework follows a modular multi-stage retrieval pipeline designed for large-scale retrieval over approximately 3.1 million documents.

Step 1 – Candidate Selection: BM25 is used as the initial lexical retriever to obtain the top-5,000 candidate documents for each query. The candidate size refers to the number of documents retrieved by BM25 before any dense retrieval or reranking stage is applied. These candidate documents form the retrieval space for subsequent neural retrieval models.

Step 2 – Neural Enhancement: Two different neural retrieval strategies are investigated in parallel:

- **Hybrid Retrieval:** Dense retrieval scores produced by SBERT and BGE-M3 are combined with BM25 scores using min-max normalization and weighted linear interpolation with coefficient λ .
- **Neural Reranking:** MonoT5 and BGE-Reranker models directly rerank the BM25 candidate documents through query-document interaction without explicit score fusion. Neural reranking experiments required substantially higher computational cost compared to lexical retrieval baselines.

Step 3 – Comparative Evaluation: All retrieval systems are evaluated using nDCG@10, nDCG@1000, Recall@10, Recall@1000, and RR@10 metrics in order to assess retrieval effectiveness at both early and deep ranking levels.

C. Experiment Tracking and Statistical Significance

Hyperparameter sweeps and experimental tracking are conducted using Weights & Biases (WANDB). In particular, λ interpolation coefficients are systematically evaluated for hybrid retrieval models. Statistical significance tests based on the Wilcoxon signed-rank test ($p < 0.05$) were performed to compare retrieval systems on a per-query basis.

IV. DATA OVERVIEW

In this study, the TREC 2024 Tip-of-the-Tongue dataset is used for an information retrieval task. It contains 3 parts: queries, qrels, and a document corpus. The aim is to find correct document from corpus based on queries. However, these queries sometimes do not contain exact keywords; instead, they are unclear and vague. It has three splits: train, dev1, and dev2. For this study, the train split is used for training, dev1 is used for validation, and dev2 is used for testing. The queries split has 150 queries, divided into two columns: query_id and query. The query_id is the unique identifier, and query is the query string. In addition, the qrels split has four columns: query_id, iteration, doc_id, and relevance. The relevance column contains relevance label, and all queries in dataset are relevant to one document. Also, corpus has 3,185,450 documents in five columns: doc_id, title, wikidata_id, text, and sections. The number of queries is small, but the corpus is very large. This is a challenging scenario

in which model has to select the right document from large corpus based on vague description. The dataset is clean and does not have any missing values or duplicate records in data quality part. Hence, we do not have to perform any data cleaning.

V. EXPLORATORY DATA ANALYSIS

The dataset is split into three equal parts. The training, validation, and testing sets have average lengths of 147.38, 139.33, and 144.30 tokens, respectively. Medians are 127.0, 118.5, and 125.5 tokens for the corresponding groups. So, it is well-balanced.

Each of the three splits contains an equal number of queries and relevance judgments (qrels), ensuring a balanced dataset partition. The average length of queries is relatively high, suggesting that queries are descriptive rather than keyword-based. The average and median lengths of queries are also close.

Semantic similarity between queries was computed using a pre-trained SBERT model (all-MiniLM-L6-v2). The average cosine similarity values were moderate and close across all splits: 0.3577 for train, 0.3717 for validation, and 0.3705 for test. This suggests that the dataset is relatively consistent across splits, while still containing semantically diverse queries. Moreover, most queries have medium length, although some longer queries exist. These observations indicate that ToT queries are generally descriptive rather than purely keyword-based, which may give semantic retrieval models such as SBERT, BGE-M3, and MonoT5 an advantage over purely lexical models such as BM25.

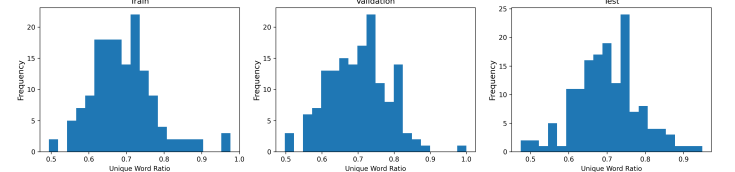


Fig. 1. Distribution of unique words across train, validation, and test splits

The relatively high unique-word ratio suggests that many queries are descriptive and lexically varied. This may make exact lexical matching harder and motivates the use of semantic retrieval models in addition to lexical baselines.

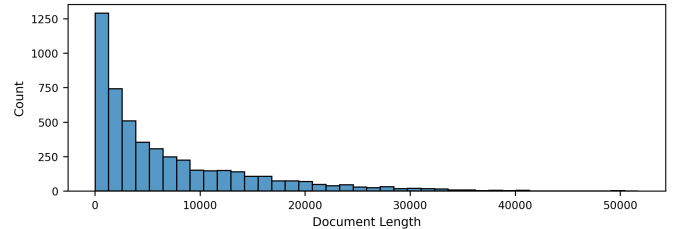


Fig. 2. Distribution of document lengths in the corpus.

Due to the large corpus size (3M+ documents), we analyze a sampled subset for efficiency. The distribution of corpus length is right-skewed. Most of the documents are of relatively short

length, while some of the documents are of extremely long length. There are some outliers in the data.

VI. CHALLENGES OF TIP-OF-THE-TONGUE RETRIEVAL TASK AND RELATED WORKS

In comparison with conventional ad-hoc retrieval queries, the TREC 2024 Tip-of-the-Tongue (ToT) retrieval task is intrinsically more difficult since the user queries are frequently vague, uncompleted, and have semantic ambiguity. Thus, state-of-the-art neural retrieval models obtain only moderate performance scores for their retrieval effectiveness. As can be seen from the TREC ToT 2024 benchmarks [5], the performance score of the BM25 model ($k_1=1$, $b=1$) is 0.0657 in terms of nDCG@10; the nDCG@10 score of the dense retrieval model based on SBERT is 0.1040, while its nDCG@1000 is 0.1665. DSGT at TREC ToT 2025 [6] studied further hybrid retrieval pipelines that combine sparse retrieval, dense retrieval, LLM-based retrieval, and reranking techniques. The researchers found out that hybrid lexical-semantic retrieval systems significantly outperform lexical-only baselines.

VII. EXPERIMENTAL SETUP AND HYPERPARAMETER TUNING

All experiments were conducted using fixed random seeds and logged through Weights & Biases in order to ensure experiment reproducibility and transparent hyperparameter tracking. The experimental results indicate that combining lexical and semantic retrieval signals produces more stable retrieval performance than using either approach independently. In particular, hybrid retrieval consistently improved nDCG@10 and deep-ranking effectiveness across multiple settings.

A. BM25 Candidate Size Analysis

Several BM25 retrieval depths were tested to find the candidate size. The recall of candidates went up from 0.1867 to 0.7667 as the number of documents retrieved grew from 50 to 20,000. nDCG@10 stayed the same at 0.0805 for all candidate sizes. This shows that the depth of retrieval did not affect the ranked BM25 documents.

The recall of candidates reached 0.6067 at a retrieval depth of 5,000 documents. Then it only went up a little to 0.6867 and 0.7667 at depths of 10,000 and 20,000 documents. For retrieval and neural reranking models a balance between coverage and efficiency led to choosing 5,000 documents as the candidate size. Recall@k and candidate recall are not the same.

Before neural retrieval is used Recall@k calculates the percent of documents returned from the BM25 candidate set. Candidate recall sets the limit of retrieval performance for downstream dense retrieval and reranking models. Therefore, candidate recall can be interpreted as an upper bound for any downstream retrieval architecture operating on the BM25 candidate pool. This is because they can only work with documents in the candidate pool. So maximizing candidate recall is key to making sure relevant documents are available for retrieval phases. BM25 candidate recall is crucial, for retrieval phases. The BM25 candidate size directly impacts

the performance of models. Therefore the choice of 5,000 documents as the candidate size is a trade-off.

This finding shows how crucial the quality of candidate generation is in multi-stage retrieval systems. Because all later dense retrieval and reranking methods work with the BM25 pool, their effectiveness is limited by the initial recall. So, the system can only be as good as its first stage allows.

B. BM25 Parameter Tuning

We chose the candidate retrieval depth and then used Weights & Biases to tune the BM25 hyperparameters. While checking k_1 and b across different values, the combo of $k_1 = 0.9$ and $b = 0.75$ gave the best results. This setup boosted Recall@10 to 0.1067, RR@10 to 0.0721, nDCG@10 to 0.0808, nDCG@100 to 0.1054, nDCG@1000 to 0.1281, and candidate recall to 0.6133. This configuration was therefore selected for all subsequent experiments.

Also, the results showed candidate recall being more sensitive to parameter changes than ranking metrics. This suggests BM25 hyperparameters mostly affected how many candidates were covered, not the order of top-ranked documents.

C. Hybrid Retrieval Weight Optimization

A preliminary study used 40 queries and 500 documents to pick the best max sequence length. For SBERT, the top scores came with a length of 256 – that’s an nDCG@10 of 0.0309, RR@10 of 0.0261, and Recall@10 of 0.0467. BGE-M3, however, hit its stride at 384, getting an nDCG@10 of 0.0335, RR@10 of 0.0293, and Recall@10 of 0.0467.

To run hybrid retrieval tests, another max sequence length study was done on those same two models. It looked at lengths of 256, 384, and 512, but this time with just 500 documents. The results SBERT topped its game at 256, with the highest nDCG@10 score. On the other hand, BGE-M3 shined brightest at 384.

The hybrid search approach used BM25 and dense retrieval scores normalized using min-max normalization with weighted linear interpolation:

$$Score(q, d) = (1 - \lambda) \cdot BM25(q, d) + \lambda \cdot Dense(q, d) \quad (1)$$

where $\lambda \in [0, 1]$ controls the contribution of lexical and semantic retrieval signals.

If $\lambda = 0$, then the method would be the one for BM25 search, while $\lambda = 1$ would stand for the pure dense retrieval model.

Hybrid retrieval experiments were carried out using the chosen maximum lengths of sequences in each case using a candidate pool of 5,000 documents. Values for the interpolation parameter λ ranging from 0.0 to 1.0 were tested at intervals of 0.1 to find the best compromise between the two retrieval signals.

From the results below, it can be seen that hybrid ranking was superior compared to either pure BM25 or pure dense retrieval ranking. In the case of SBERT, the best result was achieved at $\lambda = 0.7$ with an nDCG@10 of 0.1499. For BGE-M3, the best nDCG@10 score of 0.1559 was achieved when $\lambda = 0.8$.

TABLE I
HYBRID RETRIEVAL WEIGHT OPTIMIZATION RESULTS (SBERT: 256,
BGE-M3: 384)

Model	λ	nDCG@10	nDCG@100	nDCG@1000	RR@10	Recall@10
SBERT	0.0	0.0808	0.1054	0.1281	0.0721	0.1067
SBERT	0.1	0.0851	0.1092	0.1349	0.0759	0.1133
SBERT	0.2	0.0907	0.1203	0.1450	0.0814	0.1200
SBERT	0.3	0.0973	0.1306	0.1548	0.0863	0.1333
SBERT	0.4	0.1039	0.1385	0.1640	0.0906	0.1467
SBERT	0.5	0.1259	0.1509	0.1837	0.1076	0.1867
SBERT	0.6	0.1355	0.1636	0.1930	0.1150	0.2000
SBERT	0.7	0.1499	0.1835	0.2063	0.1321	0.2067
SBERT	0.8	0.1348	0.1724	0.1959	0.1188	0.1867
SBERT	0.9	0.1115	0.1468	0.1726	0.0923	0.1733
SBERT	1.0	0.0866	0.1283	0.1553	0.0741	0.1267
BGE-M3	0.0	0.0808	0.1054	0.1281	0.0721	0.1067
BGE-M3	0.1	0.0890	0.1129	0.1398	0.0814	0.1133
BGE-M3	0.2	0.0901	0.1178	0.1465	0.0827	0.1133
BGE-M3	0.3	0.0927	0.1261	0.1548	0.0862	0.1133
BGE-M3	0.4	0.0990	0.1363	0.1676	0.0924	0.1200
BGE-M3	0.5	0.1163	0.1503	0.1812	0.1054	0.1533
BGE-M3	0.6	0.1414	0.1666	0.1952	0.1199	0.2133
BGE-M3	0.7	0.1541	0.1824	0.2062	0.1300	0.2333
BGE-M3	0.8	0.1559	0.1900	0.2115	0.1386	0.2133
BGE-M3	0.9	0.1360	0.1693	0.1924	0.1158	0.2000
BGE-M3	1.0	0.1222	0.1552	0.1778	0.1028	0.1867

This shows that it is better to combine the features of both lexical and semantic retrieval ranking when it comes to handling vague ToT questions, and that BGE-M3 was the most efficient in this aspect.

Additionally, the findings suggest that dense retrieval on its own is inadequate for comprehensively addressing the requirements of ToT queries; hybrid retrieval, on the other hand, gains advantage through lexical matching and semantics. This synergy between the two contributes to the consistent improvements seen across various measures of performance.

D. Neural Reranking Experiments

Aside from dense hybrid search, interaction-based neural rerankers were tested using BGE-Reranker and MonoT5. An interpolation approach was also used in testing the difference between reranker performance alone at $\lambda = 1$ and hybrid reranking with BM25.

In order to identify good reranker architectures, a preliminary study was run on 40 validation queries and 500 BM25 documents. There were four instructions considered for each reranker (no instruction, ToT instruction, semantic instruction, memory instruction) and three maximum token counts (256, 384, 512). In the case of ToT, the instruction to the model was to search for the best-matching document to a vague tip-of-the-tongue question. In semantic searching, the emphasis was placed on finding the semantically most relevant document for an incomplete memory-based search request. In memory retrieval, the request was treated as an incomplete memory search task with a matching question.

For BGE-Reranker, the optimal probing setting is the one with ToT instruction and a maximum length of 256, reaching an nDCG@10 score of 0.0205. For MonoT5, the best setting is achieved with semantic instructions and maximum length 384, giving an nDCG@10 score of 0.0209. These settings will be utilized for the following reranking experiment with 5,000 candidates.

TABLE II
INSTRUCTION AND MAXIMUM LENGTH PROBE RESULTS FOR MONOT5
AND BGE-RERANKER

Model	Instruction	Max Length	nDCG@10
MonoT5	None	256	0.0119
MonoT5	None	384	0.0086
MonoT5	None	512	0.0086
MonoT5	ToT	256	0.0124
MonoT5	ToT	384	0.0118
MonoT5	ToT	512	0.0101
MonoT5	Semantic	256	0.0104
MonoT5	Semantic	384	0.0209
MonoT5	Semantic	512	0.0184
MonoT5	Memory	256	0.0159
MonoT5	Memory	384	0.0152
MonoT5	Memory	512	0.0155
BGE-Reranker	None	256	0.0143
BGE-Reranker	None	384	0.0114
BGE-Reranker	None	512	0.0114
BGE-Reranker	ToT	256	0.0205
BGE-Reranker	ToT	384	0.0134
BGE-Reranker	ToT	512	0.0113
BGE-Reranker	Semantic	256	0.0182
BGE-Reranker	Semantic	384	0.0125
BGE-Reranker	Semantic	512	0.0099
BGE-Reranker	Memory	256	0.0173
BGE-Reranker	Memory	384	0.0097
BGE-Reranker	Memory	512	0.0097

Among all retrieval models considered for evaluation, the most successful configuration turned out to be BGE-M3, with $\lambda = 0.8$ being equal to an nDCG@10 of 0.1559 and nDCG@1000 of 0.2115. As for SBERT, its performance peaked when $\lambda = 0.7$ (nDCG@10 of 0.1499).

It can be clearly observed that the combination of lexical BM25 and dense retrieval signals generally produced stronger retrieval effectiveness than using either technique independently. This observation is consistent across both SBERT and BGE-M3 experiments, suggesting that lexical matching and semantic similarity provide complementary retrieval signals for Tip-of-the-Tongue queries.

Optimal configurations of both SBERT and BGE-M3 models were obtained by means of hybrid weighting (not by purely dense retrieval ($\lambda = 1$) or purely BM25 retrieval ($\lambda = 0$)). This finding shows that just using semantic retrieval wasn't enough for the tip-of-the-tongue task. The best results came when they combined lexical and semantic evidence through score fusion, though.

For the neural reranking approach as can be seen below, BGE-Reranker showed itself to be superior to MonoT5. The best performing BGE-Reranker configuration included the ToT instruction, max length of 256, and $\lambda = 0.5$, resulting in an nDCG@10 of 0.1207. The best MonoT5 configuration involved the semantic instruction, a max length of 384, and $\lambda = 0.1$, providing an nDCG@10.

Overall, the best-performing hybrid retrieval configurations achieved higher nDCG@10 scores than the evaluated neural reranking configurations. It can be concluded that for imprecise Tip-of-the-Tongue queries, the combination of lexical and dense retrieval features performs better than using neural reranking on top of BM25 results.

TABLE III
NEURAL RERANKING WEIGHT OPTIMIZATION RESULTS
(BGE-RERANKER: ToT, 256; MONOT5: SEMANTIC, 384)

Model	λ	nDCG@10	nDCG@100	nDCG@1000	RR@10	Recall@10
BGE-RR	0.0	0.0808	0.1054	0.1281	0.0721	0.1067
BGE-RR	0.1	0.0876	0.1135	0.1362	0.0794	0.1133
BGE-RR	0.2	0.0880	0.1184	0.1402	0.0797	0.1133
BGE-RR	0.3	0.0911	0.1229	0.1462	0.0818	0.1200
BGE-RR	0.4	0.0995	0.1322	0.1549	0.0885	0.1333
BGE-RR	0.5	0.1207	0.1453	0.1683	0.1048	0.1733
BGE-RR	0.6	0.1121	0.1372	0.1616	0.0935	0.1733
BGE-RR	0.7	0.0942	0.1228	0.1438	0.0712	0.1733
BGE-RR	0.8	0.0733	0.1078	0.1298	0.0591	0.1200
BGE-RR	0.9	0.0579	0.0815	0.1121	0.0466	0.0933
BGE-RR	1.0	0.0432	0.0637	0.0912	0.0317	0.0800
MonoT5	0.0	0.0808	0.1054	0.1281	0.0721	0.1067
MonoT5	0.1	0.0820	0.1063	0.1291	0.0721	0.1133
MonoT5	0.2	0.0798	0.1087	0.1306	0.0711	0.1067
MonoT5	0.3	0.0808	0.1080	0.1332	0.0703	0.1133
MonoT5	0.4	0.0809	0.1081	0.1330	0.0686	0.1200
MonoT5	0.5	0.0649	0.1001	0.1239	0.0559	0.0933
MonoT5	0.6	0.0511	0.0865	0.1097	0.0422	0.0800
MonoT5	0.7	0.0483	0.0770	0.1011	0.0385	0.0800
MonoT5	0.8	0.0339	0.0600	0.0857	0.0260	0.0600
MonoT5	0.9	0.0149	0.0380	0.0626	0.0115	0.0267
MonoT5	1.0	0.0104	0.0210	0.0466	0.0074	0.0200

E. Statistical Significance Analysis

TABLE IV
STATISTICAL SIGNIFICANCE ANALYSIS ON NDCG@10

Comparison	Mean A	Mean B	Diff (%)	p-value	Sig.
Base SBERT vs Hybrid SBERT	0.0866	0.1499	73.23	0.0237	True
Base BGE-M3 vs Hybrid BGE-M3	0.1222	0.1559	27.55	0.1487	False
Base Reranker vs Hybrid Reranker	0.0432	0.1207	179.63	0.0581	False
Base MonoT5 vs Hybrid MonoT5	0.0104	0.0820	686.82	0.0421	True
BM25 vs Hybrid Reranker	0.0805	0.1207	50.03	0.0535	False
BM25 vs Hybrid MonoT5	0.0805	0.0820	1.88	0.9958	False
BM25 vs Hybrid SBERT	0.0805	0.1499	86.34	0.0217	True
BM25 vs Hybrid BGE-M3	0.0805	0.1559	93.73	0.0177	True
Hybrid BGE-M3 vs Hybrid MonoT5	0.1559	0.0820	-47.41	0.0261	True
Hybrid Reranker vs Hybrid MonoT5	0.1207	0.0820	-32.09	0.0383	True
Hybrid SBERT vs Hybrid MonoT5	0.1499	0.0820	-45.32	0.0230	True
Hybrid SBERT vs Hybrid BGE-M3	0.1499	0.1559	3.96	0.7118	False

Table IV shows the statistical significance testing results based on the nDCG@10 metric. The findings indicate that hybrid retrieval approaches generally outperform their corresponding baseline models. For example, Hybrid SBERT achieved an improvement of 86.34% over BM25 and was found to be statistically significant ($p < 0.05$). Similarly, Hybrid BGE-M3 improved nDCG@10 by 93.73% compared to BM25 and was also statistically significant. In contrast, the improvements achieved by the hybrid reranking approaches were not statistically significant at the 0.05 level.

Out of all the systems evaluated, the highest retrieval effectiveness was achieved by Hybrid BGE-M3. However, the performance difference between Hybrid BGE-M3 and Hybrid SBERT was not statistically significant ($p = 0.7118$), indicating that both hybrid dense retrieval approaches performed similarly on the ToT retrieval task.

Finally, MonoT5 exhibited weaker performance than the other hybrid retrieval models. While the MonoT5 approach proved highly effective in comparison with the base-line method of MonoT5, its efficiency was still significantly infe-

rior compared to the hybrid methods, namely SBERT, BGE-M3, and BGE-Reranker.

VIII. ERROR ANALYSIS

In total, three different types of behavior have been observed. Firstly, hybrid search proved itself to be efficient for handling lexical mismatches. As an illustration, the query “george clooney movie” resulted in a significant improvement compared to BM25 due to the ability of the dense model to detect semantically related documents without sufficient lexical cues.

Secondly, there were instances when BM25 performed better than the hybrid model. The example would be the description of *Sucker Punch* (2011) movie. Here, the lexical retrieval was able to correctly rank the target document, while the semantic search returned incorrect movies like *Girlhood* and *Party Girl* since they had similar semantic structure. In other words, at times, semantic search could lead to mis-ranking of the document since several documents had similar themes.

Thirdly, neither BM25 nor hybrid retrieval worked well with extremely ambiguous Tip-of-the-Tongue queries. Queries like “scary movie set in louisiana” and “chased by cops, saved by truckers” lacked sufficient identifying information for both approaches.

Therefore, the study showed that lexical and semantic approaches have their respective advantages. Lexical search handles lexical mismatches better, whereas BM25 works well when there are distinct keywords. However, both models still suffer when it comes to Tip-of-the-Tongue queries.

Overall, the observed errors indicate that retrieval effectiveness depends on both lexical evidence and semantic cues. Queries lacking both types of signals remain challenging for all evaluated approaches.

IX. INTERPRETABILITY ANALYSIS

Top-1 Agreement Analysis shows significant discrepancies in ranking between lexical and hybrid approaches. For example, the agreement between BM25 and Hybrid SBERT on top-ranking document is achieved only in 21.33% of queries, which means that the document ranked at number one position changes in nearly 80% of cases. In a similar way, the agreement between Hybrid SBERT and Hybrid BGE-M3 is obtained in just 32.67% of queries. The results clearly show that dense retrieval features play an important role in ranking process and often lead to selecting new documents that would not be chosen otherwise based on lexical matching.

Furthermore, an outcome-based evaluation at the query level was done to make a comparison between Hybrid SBERT and BM25. It has been found that Hybrid SBERT had better performance in terms of nDCG@10 compared to BM25 in 13.33% of the total number of queries, whereas in just 2.67% of queries, BM25 performed better than Hybrid SBERT. In the rest of the queries, both methods performed equally.

These findings suggest that hybrid retrieval does not simply improve existing BM25 rankings but frequently introduces

different documents into top-ranked positions. This behavior indicates that semantic retrieval signals contribute complementary information beyond lexical matching.

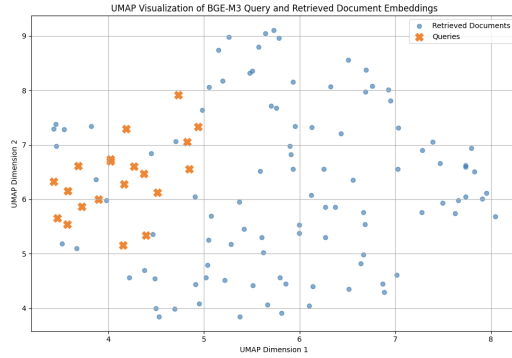


Fig. 3. UMAP projection of BGE-M3 embeddings

The two-dimensional UMAP projection of the BGE-M3 embedding space is shown in Fig. 3 above. The query embeddings cluster together into a relatively compact area, while the retrieved documents scatter more broadly within the space. While there are no distinguishable clusters, the representation hints at the ability to capture semantic meaning through dense retrieval. Hence, this UMAP visualization can only be taken as indicative proof of semantic matching and not a concrete explanation of the retrieval process.

X. FUTURE WORKS

Due to computational limitations, such LLM-based approaches were not explored in this study. While there is an obvious improvement in the proposed approach compared to the baseline model, there is much more that needs to be explored. For example, future studies might explore larger models that have been fine-tuned on instructions for semantic matching for fuzzy queries associated with the Tip-of-the-Tongue problem. Moreover, future work may also consider fine-tuning the model with supervised training using a ToT-oriented dataset.

Besides, future studies may look into LLM-based ranking of search results since LLMs will be able to perform longer reasoning and thus resolve semantic ambiguities, which remain a challenge for both lexical and dense models. Additionally, one should consider alternative approaches to hybridization and ranking based on learning-to-rank algorithms instead of linear interpolation.

XI. COMPUTATIONAL LIMITATIONS

Due to limitations in the computational capacity and the large size of the TREC ToT dataset (around 3.1 million documents), dense retrieval and neural reranking tasks were conducted using candidate sets generated via BM25, as opposed to the entire corpus. In doing so, considerable savings in computational costs have been realized without compromising the retrieval pipeline.

Nevertheless, such an approach comes with limitations, as documents that fail to be retrieved by the BM25 candidate

generation model will not be able to be recovered via any further retrieval models downstream. Effectively, the success rate of other retrieval pipelines is inextricably linked to the performance of the first BM25 phase.

In addition, LLM-based reranking at a large scale was not computationally viable. Hence, this paper only explored methods regarding dense retrieval, hybrid retrieval, and neural reranking.

XII. CONCLUSION

In this research, different neural retrieval techniques have been examined to be used for the retrieval phase of the ToT task. The performance analysis found that the strongest hybrid retrieval configurations consistently achieved higher retrieval effectiveness than the standalone lexical and dense retrieval baselines. Hybrid BGE-M3 achieved the best retrieval performance among all the tested retrieval methods and reached an nDCG@10 score of 0.1559, closely followed by Hybrid SBERT at 0.1499. Furthermore, statistical significance test showed that both Hybrid SBERT and Hybrid BGE-M3 achieved significant performance improvement over the BM25 base model. The interpretability and error analyses suggested that hybrid retrieval techniques can benefit from leveraging lexical and semantic signal to deal with ambiguous and non-specific ToT queries. Nevertheless, lexical and semantic retrieval were still challenging with highly ambiguous queries which have not provided sufficient identifying information.

Overall, hybrid retrieval was found to be an effective technique for improving retrieval performance on the Tip-of-the-Tongue task and mitigating lexical mismatch issues.

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Code Availability: The implementation of this project is available at:

<https://github.com/Gozde123/trec-tot-retrieval>

W&B Availability: The experiment logs and reports are available at Interim 1 W&B Report and Interim 2 W&B Report. Also the final report can be seen WB Reports: Final Tables W&B Report, Final Plots W&B Report.